

CCSU BCA 1st Semester – Mathematical Foundation for Computer Science-I

2026 Exam-Oriented Important Notes

These notes are a compact revision guide. The CCSU official site lists BCA 2025–26 under the Department of Computer Application. Match these topics with your college's current syllabus before the exam.

1. Sets

- Set: A well-defined collection of distinct objects. Example: $A = \{1, 2, 3\}$.
- Types: empty/null set, singleton set, finite set, infinite set, equal sets, equivalent sets, subset and proper subset.
- Operations: Union $A \cup B$, Intersection $A \cap B$, Difference $A - B$, Complement A' .
- Important formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- Power set $P(A)$: set of all subsets of A . If $n(A) = n$, then $n(P(A)) = 2^n$.

2. Relations

- Relation from A to B is a subset of $A \times B$.
- Important terms: domain, range, reflexive, symmetric, antisymmetric and transitive.
- Equivalence relation: reflexive + symmetric + transitive.
- Identity relation: every element is related to itself.

3. Functions

- Function $f: A \rightarrow B$ assigns every element of A exactly one element of B .
- One-one (injective): different inputs have different outputs.
- Many-one: two or more inputs may have the same output.
- Onto (surjective): every element of codomain has a pre-image.
- Bijective: both one-one and onto; a bijection has an inverse function.

4. Matrices

- Matrix: rectangular arrangement of numbers in rows and columns.
- Types: row, column, square, diagonal, scalar, identity, zero, symmetric and skew-symmetric.
- Matrix operations: addition, subtraction and multiplication.
- Determinant of $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.
- Inverse of A exists when $|A| \neq 0$. For a 2×2 matrix, $A^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
- Cramer's Rule is used to solve simultaneous linear equations.

5. Permutation & Combination

- Permutation: arrangement where order matters. ${}^nP_r = \frac{n!}{(n-r)!}$
- Combination: selection where order does not matter. ${}^nC_r = \frac{n!}{r!(n-r)!}$.
- Relation: ${}^nP_r = {}^nC_r \times r!$.
- Important identity: ${}^nC_r = {}^nC_{n-r}$.

- Practice numerical questions involving arrangements, selections and repeated objects.

6. Mathematical Induction

- Principle: prove a statement for $n=1$ (or initial value), then assume it true for $n=k$ and prove it for $n=k+1$.
- Two main steps: Base case and Inductive step.
- Common questions: sums of natural numbers, squares/cubes and divisibility.

7. Differential Calculus – Revision

- Derivative represents the rate of change of a function.
- Basic rules: $d(x^n)/dx = nx^{n-1}$; $d(\sin x)/dx = \cos x$; $d(\cos x)/dx = -\sin x$; $d(e^x)/dx = e^x$.
- Product rule: $(uv)' = u'v + uv'$.
- Quotient rule: $(u/v)' = (vu' - uv')/v^2$.
- Chain rule: $d[f(g(x))]/dx = f'(g(x))g'(x)$.
- Practice derivatives of algebraic, trigonometric, exponential and composite functions.

8. Integral Calculus – Revision

- Integration is the reverse process of differentiation.
- $\int x^n dx = x^{n+1}/(n+1) + C$, $n \neq -1$.
- $\int 1/x dx = \ln|x| + C$.
- $\int e^x dx = e^x + C$; $\int \sin x dx = -\cos x + C$; $\int \cos x dx = \sin x + C$.
- Important methods: substitution and integration by parts.
- Practice standard indefinite and definite integrals.

■ High-Priority Exam Practice

- Set operations and Venn-diagram based numericals.
- Relations: check reflexive/symmetric/transitive and identify equivalence relations.
- Functions: identify one-one/onto/bijective and find inverse where possible.
- Matrix multiplication, determinant, inverse and simultaneous equations.
- Permutation/combination numericals.
- Mathematical induction proofs.
- Standard differentiation and integration problems.

Important Questions for 2026 Revision

1. Define a set and explain different types of sets with examples.
2. Prove/use the formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
3. Define relation. Explain reflexive, symmetric and transitive relations.
4. What is an equivalence relation? Give an example.
5. Define one-one, onto and bijective functions.
6. Find the determinant and inverse of a 2×2 matrix.
7. Solve simultaneous linear equations using Cramer's Rule.
8. Explain permutation and combination with formulas and examples.
9. State and apply the principle of mathematical induction.
10. Differentiate algebraic, trigonometric and composite functions.
11. Explain product, quotient and chain rules.
12. Evaluate standard indefinite/definite integrals.

Note: This is a revision-focused notes PDF, not a guarantee of the exact 2026 question paper. Always follow the latest syllabus/notification issued by CCSU and your college.